MIL-T-10652A(EL)
8 August 1967
SUPERSEDING
MIL-T-10652(SigC)
6 November 1950

MILITARY SPECIFICATION

TUBING, PLASTIC

GLASS FABRIC BASE, LOW PRESSURE LAMINATED

1. SCOPE

1.1 Scope. This specification covers one type of laminated glass fabric base tubing, low pressure molded used in construction of antenna masts (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-R-7575 - Resin, Polyester, Low Pressure Laminating.

STANDARDS

FEDERAL

FED-STD-1 - Standard for Laboratory Atmosphere Conditions for Testing
FED-STD-595 - Color

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 9330

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3. REQUIREMENTS

3.1 Material. The material shall be as specified herein.

3.2 Fabric. The fabric shall be a continuous-filament type glass fabric, cleaned and then treated with a finish which will produce the characteristics required by finished glass cloth, including the required performance characteristics when tested with the applicable laminating resin designated in this specification. The fabric shall be one of the following, the heavier weight fabric to be used for the thicker tubings.

COMMERCIAL NUMBER	WEAVE	WEIGHT (OZ/SQ YD)	AVERAGE THICKNESS (INCH)
181	#8 Harness Satin	8.90	0.0085
182	#8 Harness Satin	12.40	0.0130
183	#8 Harness Satin	16.75	0.0180
184	#8 Harness Satin	25.90	0.0270

3.2.1 Surface Defects. The glass fabric both before and after impregnation with resin shall be of uniform color, shall not be brittle, fused, streaked or splotchy in appearance, and shall be free from creases, wrinkles, oil-or grease-spots or other contaminations.

3.3 Resin. The resin used to impregnate the glass fabric shall be polyester conforming to grade B, Class 1 of Specification MIL-R-7575.

3.4 Foreign Materials. No metal staples, tape or other foreign materials shall remain in the finished tubing.

3.5 Overlays. The tubing shall not be made with a gel resin overlay, integrally molded or otherwise, or with any other such thick resin surface or any other surfacing material.

3.6 Machinability. The material shall be such that it can be drilled, tapped, sawed and machined in all directions in accordance with the manufacturer's recommended technique without cracking, splitting, or otherwise impairing the material for general use.

3.7 Color. Unless otherwise specified (see 6.2), color of the material shall be olive drab number X24087 in accordance with Federal Standard 595. The pigment used to color the impregnating resin shall not adversely affect the physical properties of the finished tubing.

3.8 Surface Finish. The surface finish of the tubing shall be semi-gloss as produced by the laminating process.

3.9 Dimensions. The diameter, wall thickness, length, taper if any, and the allowable tolerances therefor shall be as specified in drawings, specifications or other requirements set forth in the contract or order for the tubing (see 6.2).

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3.10 Water absorption. When tested as specified in 4.5.2, the tubing shall withstand immersion for 24 hours without showing water absorption in excess of 1 percent.

3.11 Resin content. The resin content of the tubing, when determined as prescribed in 4.5.3, shall be not less than 35 percent nor more than 47 percent.

3.12 Compressive strength. Tubing shall be tested for compressive strength as specified in 4.5.4. Tubing having a diameter less than 2-1/2 inches and a wall thickness less than 3/16 inch shall show a minimum compressive strength of 30,000 pounds per square inch; after conditioning by immersion in distilled water for 240 hours at 23 ± 2C(73.4 ± 3.6 F), the tubing shall show a minimum compressive strength of 22,000 pounds per square inch. Tubing of any diameter having a wall thickness greater than 3/16 inch or tubes having a diameter greater than 2-1/2" shall show a minimum compressive strength of 18,000 pounds per square inch; after conditioning by immersion in distilled water for 240 hours at 23 ± 2C(73.4 ± 3.6F), the tubing shall show a minimum compressive strength of 16,000 pounds per square inch.

3.13 Workmanship. The tubing shall be uniform, smooth, and free from gaps, cracks, holes, blisters, resin pockets, wrinkles, starved areas, delamination, air holes, and other defects affecting serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government.

4.1.1.1 Additional inspection. Nothing specified herein shall preclude the supplier from taking such additional samples and making such additional inspection as he may deem necessary or desirable to assure conformance of the glass-fabric tubings to this specification.

4.2 Classification of inspection. The examination and testing of the glass-fabric tubings shall be classified as follows:

(a) Quality conformance inspection (see 4.4)

1. Inspection of product for delivery (see 4.4.1).

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4.3 Inspection conditions. Unless otherwise specified herein, all inspection shall be made at 25° ± 5°C, and at room ambient relative humidity and pressure. In case of dispute, the referee conditions shall be as specified in Federal Standard No. 1.

4.4 Quality conformance inspection.

4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A, B, and C.

4.4.1.1 Inspection lot. An inspection lot, shall consist of tubings of a given diameter, length, and wall thickness prepared from an impregnating operation in which the resin and filler and conditions of cure are the same and offered for delivery at one time.

4.4.1.2 Rejected lots. If an inspection lot is rejected, the supplier may replace it with a new lot, rework it to correct the defects, or screen out the defective units, as applicable, and reinspect. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots. Rejected lots shall be inspected, using tightened inspection.

4.4.1.3 Group A inspection. Group A inspection, including sampling shall conform to Table I and general inspection level I of MIL-STD-105. Unless otherwise specified, normal inspection shall be used at the start of the contract.

TABLE I INSPECTION PROCEDURES

INSPECTION	REF. PAR.	INSP. PAR.	GROUP A(QPL)	GROUP B	GROUP C
Visual & Mechanical Major	3.2.1, 3.4, 3.6, 3.7, 3.8 & 3.12	4.5.1	1.0%	--	--
Visual & Mechanical Minor	same as above	4.5.1	4.0%	--	--
Water Absorption	3.10	4.5.2	--	See 4.4.2	--
Resin Content	3.11	4.5.3	--	4.4.2	
Compressive Strength	3.12	4.5.4	--	--	See 4.4.3

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4.4.2 Group B inspection. Group B inspection, including sampling shall conform to Table I and to MIL-STD-105. The AQL shall be 6.5 percent defective, and the special inspection level shall be S-4 for normal, reduced, and tightened inspection. Normal inspection shall be used at the start of the contract.

4.4.3 Group C inspection. Group C inspection shall conform to Table I.

4.4.3.1 Sampling for group C inspection. Ten tubings shall be selected at random each month for inspection.

4.4.3.2 Noncompliance. If a sample unit fails group C inspection, the supplier shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and on all units of product which were manufactured with the same conditions, material and processes.

4.4.5 Disposition of Group B and C samples. Samples which have been subjected to Group B and C inspection will not be shipped on the contract or order because of the destructive nature of the tests.

4.5 Methods of test. Tubings on contract or order shall be tested in accordance with test procedures hereinafter specified.

4.5.1 Visual inspection. Tubings shall be inspected for compliance with color, and workmanship requirements of this specification and the dimensional requirements of the contract or order (see 6.2). Visual and mechanical defects shall be classified by the Government as major or minor in accordance with the definitions of standard MIL-STD-105.

4.5.2 Water absorption. Two specimens in accordance with Figure 1 shall be cut from each tubing selected for test. The specimens shall be dried at 105± 2C (221 ± 3.6 F) for 1 hour ± 6 minutes, cooled to room temperature in a desiccator for at least 2 hours, and weighed. The specimens shall then be immersed in distilled water at 23 ± 2C (73.4 ± 3.6F) for not less than 24 nor more than 26 hours. They shall be removed individually as needed, all surfaces wiped off with a damp cloth, and reweighed as rapidly as consistent with accuracy within 1 minute after removal from the water. Specimens 1/16 inch or less in thickness shal be placed in a weighing bottle immediately after wiping and shall be weighed in the bottle. No correction shall be made for water-soluble matter. The percentage of water absorption of each specimen shall be calculated by use of the formula:

$$A = \frac{W_2 - W_1}{W_1} \times 100$$

Where: A = percentage of water absorption in 24 hours

W_1 = weight of specimen after drying period

W_2 = weight of specimen after immersion

The two percentages obtained from the above calculations shall be averaged. This average shall be taken as the percentage of water absorption for the tubings and shall not exceed 1 percent as prescribed in 3.10.

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4.5.3 Resin content. Two specimens in accordance with Figure 1 shall be cut from each tubing selected for test. The specimens shall be subjected to 105± 2C (221 ± 3.6 F) for 1 hour ± 6 minutes and cooled to room temperature in a dessicator for at least 2 hours. Each specimen shall be given four or five saw cuts to allow for circulation of air during ignition. All loose particles and projecting fibers shall be removed from the specimens. Each specimen shall be weighed on an analytical balance in a previously weighed crucible. Crucible and specimen shall then be ignited slowly in the flame of a Bunsen burner, in order to remove volatile matter without fusing the glass fabric. After this preliminary ignition has been completed, the crucibles and specimens shall be transferred to a muffle furnace and ignited to constant weight at 1000 to 1100 F. Higher temperatures should be avoided in order to prevent fusing of the glass fabric which would allow unburnt carbon particles to be entrapped. The ignition at 1000 to 1100 F may require from 2 to 6 hours, at the conclusion of which, the glass fabric residue should be entirely white and should show no signs of fusion. Should the carbon be incompletely removed, the residue will be various shades of gray. The residue and crucible shall be allowed to cool to room temperature in a dessicator and then reweighed on the analytical balance. The percentage of resin content of each specimen shall be calculated as follows:

$$\text{Percent Resin Content} = \frac{\text{Loss in weight of specimen}}{\text{Original weight of specimen}} \times 100$$

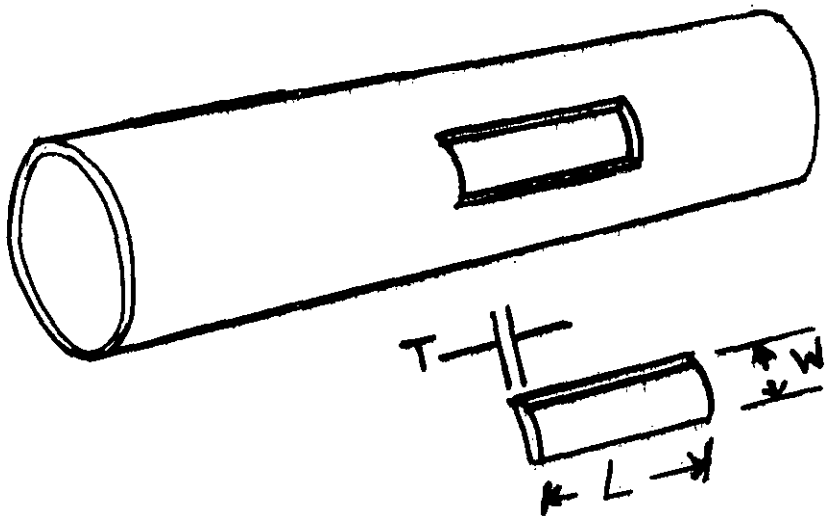
The two percentages obtained from the above calculation shall be averaged. This average shall be taken as the percentage of resin content of the tubing and shall conform to the requirements of 3.11.

4.5.4 Compressive Strength. Two specimens, prepared especially for this test in accordance with Figure 2, shall accompany each tubing selected for test. These specimens are to be made simultaneously with the production lot they are to represent and will be made under identical conditions of manufacture, with respect to materials, p[illegible]ce, equipment and procedures.

4.5.4.1 Apparatus. Any standard testing machine may be used, provided it is accurate to within one percent of the lowest load for which it is used, and provided it has parallel loading surfaces. Surface ground steel plates with parallel surfaces shall be placed between each end of the specimen and the loading heads. For purposes of safety, the top bearing plate shall be taped to the top loading head and cords shall be tied loosely around each end of the specimen and the testing machine.

4.5.4.2 Conditioning. One of the specimens shall be tested without any conditioning; the other specimen shall be conditioned by immersing in distilled water for 240 hours at 23± 2C (73.4 ± 3.6 F). It shall then be wiped dry and immediately subjected to test.

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FOR TUBINGS 1 INCH or LESS I.D.

L= 2 INCHES

W= ½ THE CIRCUMFERENCE

T= THICKNESS OF THE TUBE WALL

FOR TUBINGS OVER 1 INCH I.D.

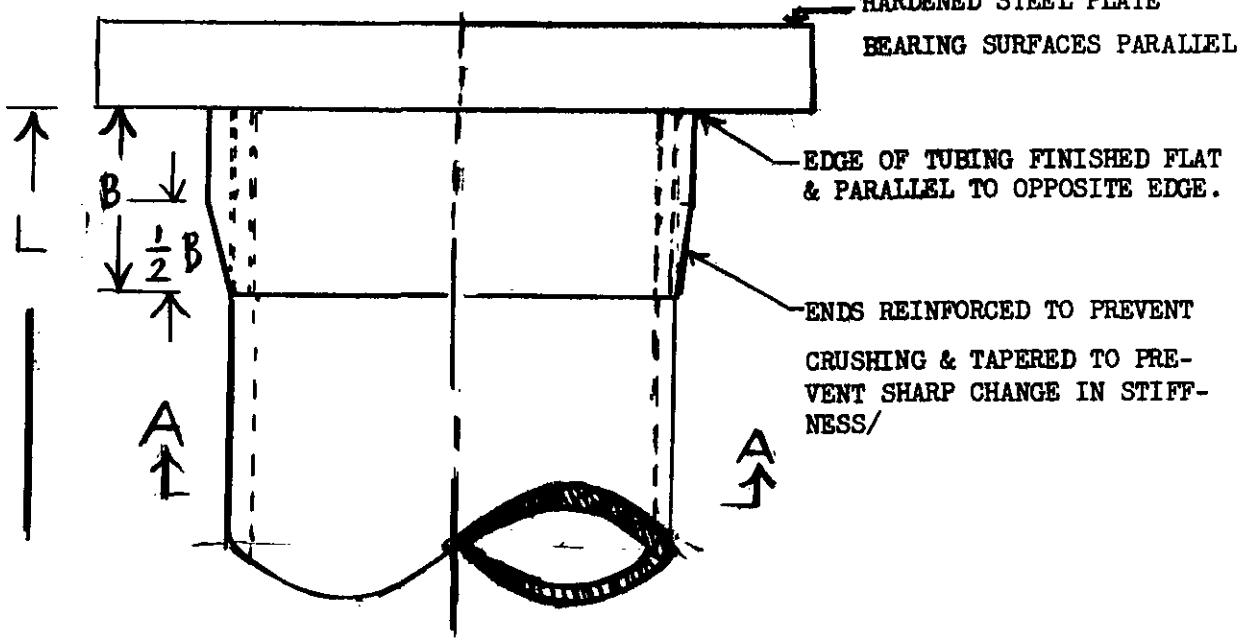
L= 2 INCHES

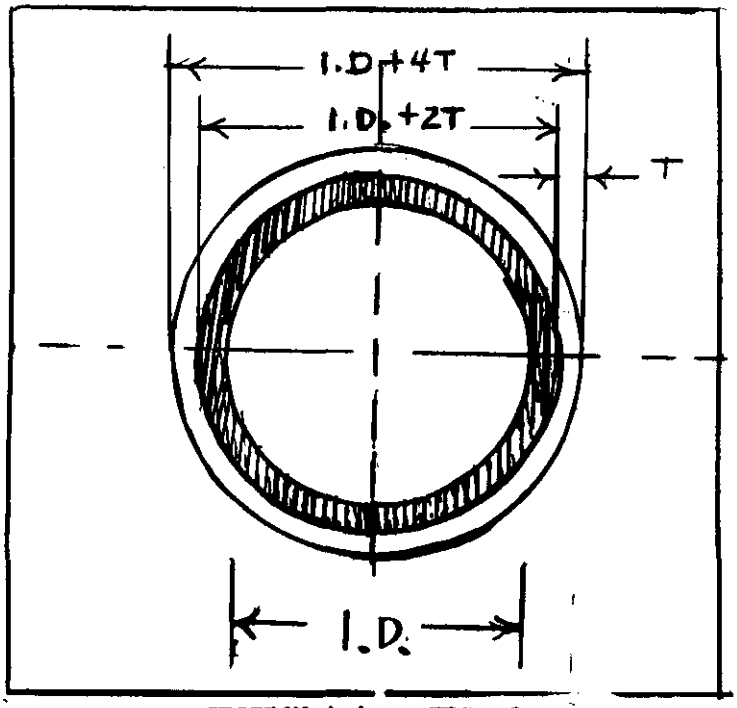
W= ½ INCH

T= THICKNESS OF THE TUBE WALL

FIG. I

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T = THICKNESS

THE LENGTH "L" IS EQUAL TO THE ID.

B = ½ inch or 1/3 ID

WHICHEVER IS SMALLER

SECTION A-A FIG. 2

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4.5.4.3 Procedure. The inside diameter and the wall thickness shall be measu to the nearest 0.001 inch. The average of at least eight measurements shall be taken. The specimen shall be carefully centered in the testing machines, and the hardened steel bearing plates so placed as to avoid eccentric loading on the specimen. Compressive load shall be applied so that the deformation does not exceed 0.010 inch per inch of length of specimen per minute, or that the stress rate does not exceed 25,000 psi per minute.

4.5.4.4 Calculations. The ultimate compressive strength in pounds per square inch shall be calculated, using the following formula:

$$S = \frac{P}{3.1416 + (d + t)}$$

Where: S= compressive strength in pounds per square inch
P= maximum load in pounds
t= wall thickness in inches
d= inside diameter in inches

The calculated compressive strength of the tubing shall conform to the requirements of 3.12.

5. PREPARATION FOR DELIVERY

5.1 This section is not applicable to this specification.

6. NOTES

6.1 Fabricated Parts or Equipment. This specification covers tubing intended primarily for use in the construction of antenna masts. Requirements applicable to such masts should be specified in separate equipment specifications or as part of contracts or orders.

6.2 Ordering Data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Dimensions required. (see 3.9)
(c) Taper if any. (see 3.9)
(d) Color if other than specified. (see 3.7)
(e) Finish if other than specified. (see 3.8)

Custodians:
Army - EL
Review:
Army - El,MR,ME
User:
Army - GL

Preparing Activity
Army - EL
Project No. 9330-A090

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and *at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in* corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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